

M. Paola Vaccaro

PhD candidate in Astrophysics

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ADS Publication List

Research interests

- Intermediate-mass black holes
- Dynamics in AGN disks
- Dynamics in star clusters
- Gravitational-wave sources
- Stellar collisions
- Einstein Telescope science

Methods and software

- **Modelling:** semi-analytical frameworks, population synthesis, SPH hydrodynamical simulations, Bayesian statistics
- **Languages:** Python, C/C++, CUDA, Fortran
- **HPC:** MPI, Slurm, GPU and cluster workflows (EuroHPC summer school, RIKEN; SSC research-software fellow)
- **Code:** lead developer of `FastClusterAGN` and `ARISE`

Languages

English (fluent, C1), Italian (native), German (basic), Spanish (basic)

Research profile

I am a computational astrophysicist and PhD student (Heidelberg University, supervisor: Prof. Michela Mapelli) modelling how black holes grow from stellar to intermediate and supermassive masses, and what their gravitational-wave mergers reveal about the early Universe. I showed that AGN disks can grow black holes through the pair-instability gap into the intermediate-mass range, but only in low-viscosity disks with efficient gas hardening, and that the resulting populations match systems like GW190521 and GW231123.

Selected highlights

- **10 papers**, of which 4 as first author. $\sim$ 80 citations on the first-author papers. h-index 7 (NASA ADS, Sept. 2026)
- **12 conference contributions** (EAS, MODEST, ET Symposium, among others) and seminars at ISTA in Vienna, GSSI in L'Aquila, the University of Arizona and UNAB in Santiago.
- **Invited** to write a contribution for a Royal Society theme issue (2025) and to give talks at the Niels Bohr Institute (2024) and the Physics of Data Workshop, Venice (2023).
- **Selected** for the SSC Fellowship (2026–27) in research software engineering, Scientific Software Center, Heidelberg; PhD fellow of IMPRS; member of the STRUCTURES Young Research Convent.
- **Mentoring and service:** co-supervised 3 students (1 MSc, 2 BSc); teaching assistant, 2 semesters; referee for A&A, JCAP and PRD; member of the Einstein Telescope collaboration (OSB Div. 3).

Scientific publications *as main author*

- **Black hole populations and hierarchical growth across the AGN parameter space**, M. P. Vaccaro, M. Mapelli, A. A. Trani, B. Liu. A&A, 714, A322 (2026)
- **The role of migration traps in the formation of binary black holes in AGN disks**, M. P. Vaccaro, Y. Seif, M. Mapelli. A&A, 708, A171 (2026)
- **Astrophysics: a modern discipline with a Newtonian origin**, M. P. Vaccaro. Phil. Trans. R. Soc. A, 383: 20230294 (2025), invited contribution to a theme issue
- **Impact of gas hardening on hierarchical black hole mergers in AGN disks**, M. P. Vaccaro, M. Mapelli, C. Perigois, et al. A&A, 685, A51 (2024)

Education

Sep. 2023 – 2028 **PhD in Astronomy and Cosmic Physics**

At Institute for Theoretical Astrophysics (ITA), ZAH, Heidelberg University in Heidelberg, Germany. Supervisor: Prof. Michela Mapelli - Thesis title: "Binary Black Hole population in the Local Universe"

Oct. 2020 – Sep. 2022 **Master degree in Astrophysics**

At Università degli Studi di Padova in Padua, Italy. Final grade: 110/110 *cum laude*, GPA: 29.4/30. Supervisor: Prof. Michela Mapelli - Thesis title: "Hierarchical Black Holes Mergers in Active Galactic Nuclei"

Sep. 2019 – Jul. 2020 **Master courses in Matter Physics**

At Università di Pisa in Pisa, Italy - GPA: 28.8/30

Sep. 2016 – Sep. 2019 **Bachelor degree in Physics**

At Università di Pisa in Pisa, Italy. Final grade: 110/110 *cum laude*, GPA: 28.6/30. Supervisor: Prof. Pier Giorgio Prada Moroni - Thesis title: "Cooling evolution of white dwarfs and cosmochronology"

Sep. 2011 – Jul. 2016 **High school diploma**

At Liceo Scientifico Alfano da Termoli in Termoli (CB), Italy. Final grade: 100/100 *cum laude*

Aug. 2014 – Jul. 2015 **High school exchange year**

At Mead High School in Mead, Colorado, USA. GPA: 4.0/4 - ACT score: 33/36

Fellowships, funding and grants

PhD in Astronomy and Cosmic Physics (2023 – 2028, expected) Heidelberg University. Funded by the European Research Council (ERC Consolidator Grant DEMOBLACK, no. 770017, PI: M. Mapelli) and by the German Excellence Strategy via the Heidelberg Cluster of Excellence STRUCTURES (EXC 2181 – 390900948).

SSC Fellowship (2026 – 2027) Selected for individual training in research software engineering at the Scientific Software Center, Heidelberg University. Competitive selection; one of **5 awardees**.

Pre-doctoral fellowship (February 1st – August 31st, 2023) in the DEMOBLACK ERC Team at Università degli Studi di Padova, Padua, Italy. Supervisor: Prof. Michela Mapelli.

Internship (October 1st – December 31st, 2022) in the GW group at IFAE-UAB, Barcelona, Spain. Supervisors: Dr. Lluïsa-Maria Mir i Martínez and Prof. Mario Martínez Pérez.

Travel grants: Research visit to the GSSI in L'Aquila, Italy, awarded through the ACME Transnational Access programme (April 2026); participation to MODEST-25 hosted in Seoul, South Korea, sponsored by the International Astronomical Union (July 2025); participation in the International HPC Summer School hosted in Kobe, Japan, sponsored by the EuroHPC Joint Undertaking via the HPC SPECTRA project (July 2024).

Memberships

Einstein Telescope collaboration (October 1st, 2023 – present), Observational Science Board (OSB), Division 3.

IMPRS (October 1st, 2023 – present) PhD fellow of the 19th generation in the International Max Planck Research School (IMPRS) for Astronomy and Cosmic Physics at Heidelberg University.

Young Research Convent (October 1st, 2023 – present) of the Heidelberg Cluster of Excellence STRUCTURES. Programme funded by the German Excellence Strategy.

Astronomers for Planet Earth (2025 – present) International movement of astronomers, working to address the climate crisis from an astronomical perspective.

Scientific publications *as co-author*

- **When black holes align: a subpopulation of aligned massive BBHs observed via GWs**, Stefano Rinaldi, Charmaine Wong, M. P. Vaccaro, et al., 2026, arXiv:2606.23305
- **The Science of the Einstein Telescope**, Adrian Abac et al. (incl. M. P. Vaccaro), 2025, arXiv:2503.12263
- **The boring history of Gaia BH3 from isolated binary evolution**, Giuliano Iorio et al. (incl. M. P. Vaccaro), A&A, 690, A144 (2024)
- **Hierarchical binary black hole mergers in globular clusters: mass function and evolution with redshift**, Stefano Torniamenti et al. (incl. M. P. Vaccaro), A&A, 688, A148 (2024)
- **GW190521: a binary black hole merger inside an active galactic nucleus?**, Sophia Morton et al. (incl. M. P. Vaccaro), Phys. Rev. D, 108, 123039 (2023)
- **The hitchhiker's guide to the galaxy catalog approach for gravitational wave cosmology**, Jonathan R. Gair et al. (incl. M. P. Vaccaro), AJ, 166, 22 (2022)

Conferences and workshops

- Contributed Talk | **Many faces of stellar-mass black holes** — Kathmandu, Nepal
  Talk title: "AGN Disks as Multi-Messenger Factories for Spinning Binary Black Holes". — October 11th–16th, 2026
- Poster Presentation | **Gamma Ray Symposium** | 31 August–4 September, 2026 — Heidelberg, Germany
- Flash Talk + Poster Presentation | **EAS annual meeting** — SwissTech Convention Centre, Lausanne, Switzerland
  Session: "SS31: Exotic populations in star clusters seen through the lens of binary interactions". Poster title: "Multi-Stage Collision Products in Dense Clusters". — 29 June–3 July, 2026
- Contributed Talk | **XVI Einstein Telescope Symposium** — RWTH Aachen University, Aachen, Germany
  Talk title: "Cosmic Evolution of Binary Black Hole Mergers from AGN Disks". — June 15th–19th, 2026
- Contributed Talk | **MODEST-25 & IAU Symposium 398** — Seoul National University, Seoul, South Korea
  Talk title: "Hierarchical Black Hole Mergers in AGN Disks". — June 16th–20th, 2025
- Contributed Talk | **Dynamix** — Institute of Astronomy, Cambridge, UK
  Title: "Hierarchical BH Mergers in AGN Disks: Tracing MBH Growth Across Cosmic Time". — June 2nd–6th, 2025
- **Invited Talk** | **New ideas on the origin of black hole mergers** — Niels Bohr Institute, Copenhagen, Denmark
  Talk title: "Specks of light from AGNs as possible indicators of the dynamical formation of binary black holes". — August 12th–16th, 2024
- Contributed Talk | **EAS annual meeting** — Congress Center, Padova, Italy
  Session: "S1: Unveiling black hole Growth across Cosmic Time in the JWST and LISA era". — July 1st–5th, 2024
  Talk title: "Hierarchical Black Hole Mergers in Active Galactic Nuclei Discs".
- Contributed Talk | **Massive Black Holes in the First Billion Years** — University College Cork, Cork, Ireland
  Talk title: "Possible constraints on the SMBH mass distribution from hierarchical BBH mergers in AGN discs". — April 29th – May 3rd, 2024
- Contributed Talk | **RESCEU-NBIA workshop on GW sources** — University of Tokyo, Tokyo, Japan
  Talk title: "The impact of Gas Hardening on Hierarchical Black Hole Mergers in Active Galactic Nuclei Disks". Contribution to the **discussion panel** "Many-Body Systems". — December 7th–15th, 2023
- Flash Talk | **Workshop "GwPopNext"** — University of Milano Bicocca, Milan, Italy
  Talk title: "Semi-analytical Models for Hierarchical Binary Black Hole Mergers". — July 10th–14th, 2023
- Poster Presentation | **Renaissance of Stellar Black-Hole Detections** — Lorentz Center, Leiden, Netherlands
  Invited participation — June 26th–30th, 2023
- **Invited Talk** | **Physics of Data Workshop** — Venetian institute of science, literature and art, Venice, Italy
  Talk title: "How do black holes grow? Active galactic nuclei as a breeding ground for intermediate-mass black holes" — April 19th–21st, 2023

Colloquia

- Galaxy Group Meeting (online) — UNAB, Santiago, Chile | August 28th, 2026
- Department Colloquium — GSSI, L'Aquila, Italy | April 24th, 2026
- Institute Seminar — ISTA, Vienna, Austria | November 13th, 2025
- "BinBoom" Group Meeting (online) — The University of Arizona | September 30th, 2025
- Physics of Stellar Objects Group Meeting — HITS, Heidelberg, Germany | April 16th, 2025
- Gravitation and Cosmology Group Meeting — GSSI, L'Aquila, Italy | December 9th, 2024
- ITP Cosmology Seminar — ITP, Heidelberg, Germany | October 29th, 2024
- Einstein Telescope monthly meeting (online) — ET-OSB Div3 | October 25th, 2024
- "Astrosticini" Seminar (online) — GSSI, L'Aquila, Italy | April 26th, 2024
- Galaxy Group Meeting — ITA-ZAH, Heidelberg, Germany | March 27th, 2024
- Planet Formation Group Meeting — ITA-ZAH, Heidelberg, Germany | February 20th, 2024

Summer Schools attendance

- "Cosmic Multiplicity: Evolution and Fate of Binary Stars" Summer School hosted by IMPRS in Heidelberg, Germany — September 14th-18th, 2026
- "AI for Astronomy" Summer School hosted by IMPRS in Heidelberg, Germany — September 8th-12th, 2025
- International HPC Summer School hosted by the RIKEN Center in Kobe, Japan — July 7th-12th, 2024
- "ML4ASTRO, Machine Learning in Astrophysics" workshop hosted by HITS in Heidelberg — June 2024; July 2025
- GW Open Data Workshop hosted by GWOSC (online event) — May 23rd-25th, 2022

Teaching experience and academic service

- Reviewer for *Astronomy & Astrophysics* (*A&A*), *Journal of Cosmology and Astroparticle Physics* (*JCAP*), *Physical Review D* (*PRD*) for a total of 5 papers. — since 2025
- Teaching assistant for "Introduction to Astronomy and Astrophysics": Conducted daily exercise tutorials, corrected student submissions, and provided detailed feedback. — Summer Semesters 2024, 2025
- Student co-supervision
  - Master student Julian L. Saling, "Hydrodynamical simulations of massive-star collisions" — 2025/2026
  - Bachelor student Yannick Seif, "Pair Up of Binary Black Holes in Active Galactic Nuclei" — 2024/2025
  - Bachelor student Jona A. Schubert, "The mass ratio of binary black holes: impact of mass transfer" — 2024
- Volunteer staff member at EAS Annual Meetings — July 2024, July 2026

Outreach and Media Coverage

- Research featured in Gist.Science: "The formation of binary black holes in AGN disks". — April 8th, 2026
- Author of "The Origins of Black Hole Mergers: Diverse Pathways Across the Universe", blog post for the STRUCTURES Cluster of Excellence, aimed at a general audience. — March 7th, 2025
- Co-organizer for the workshop "A dance of stars and black holes" of the Girls' Day program "Physik hautnah!" for school-aged students — 2024, 2025, 2026
- Member of the organizing committee for the Scholé "School of Physics" for high schoolers — since 2022

Last updated: November, 2026.